

AI Enabled IoT Smart Energy Monitoring and Home Automation System

1. Professional Project Title

AI Enabled IoT Smart Energy Monitoring and Home Automation System using ESP32, Blynk, and Google Assistant

Alternative Titles:

- * Smart Home Automation and Energy Monitoring System using ESP32
- * IoT Based Smart EMS with Voice Control
- * Embedded IoT Energy Monitoring and Automation System
- * Wireless Smart Home Control and Energy Analytics System

2. Aim of the Project

The aim of this project is to design and develop an ESP32-based IoT Smart Energy Monitoring and Home Automation System capable of monitoring electrical parameters such as voltage, current, and power while enabling wireless appliance control using Blynk IoT and Google Assistant through Sinric Pro.

3. Abstract

This project presents an AI-enabled IoT Smart Energy Monitoring and Home Automation System using ESP32. The system integrates energy monitoring, relay-based automation, wireless communication, and cloud IoT technology into a single embedded platform.

The proposed system measures electrical parameters including voltage, current, and power using ACS712 and ZMPT101B sensors. The measured data is displayed on an OLED display and transmitted to the Blynk IoT cloud platform for remote monitoring and analytics.

The system also supports smart appliance automation using relay modules to control devices such as bulbs and fans. Google Assistant voice control is implemented through Sinric Pro, enabling users to operate appliances using voice commands.

This project combines embedded systems, IoT, automation, wireless communication, cloud monitoring, and smart home technologies into a professional engineering solution suitable for smart homes, industries, and energy-saving applications.

4. Problem Statement

Traditional electrical systems do not provide real-time energy monitoring, remote accessibility, or intelligent automation. Users are unable to monitor energy consumption remotely or control appliances efficiently.

Manual switching leads to unnecessary energy wastage and reduced operational efficiency.

Therefore, there is a need for a smart, wireless, and IoT-based system capable of:

- * Real-time energy monitoring
- * Wireless appliance control
- * Voice assistant integration

- * Cloud-based monitoring
- * Smart automation
- * Energy-saving management

5. Objectives

- * To develop a real-time energy monitoring system
- * To measure voltage, current, and power consumption
- * To implement wireless IoT monitoring using Blynk
- * To control appliances remotely using ESP32
- * To integrate Google Assistant voice control
- * To automate home appliances using relays
- * To display real-time data on OLED display
- * To create a professional embedded IoT system

6. Core Concept

The ESP32 acts as the central controller of the system.

The ACS712 current sensor and ZMPT101B voltage sensor measure electrical parameters from connected loads.

The ESP32 processes sensor data and calculates:

- * Voltage
- * Current
- * Power

The calculated values are:

- * Displayed on OLED
- * Sent to Blynk Cloud
- * Visualized using SuperChart

Relay modules are used to control appliances such as bulbs and fans.

Sinric Pro connects the ESP32 with Google Assistant for voice-based automation.

7. Components Required

Component	Quantity	Purpose
ESP32 Dev Board	1	Main controller
ACS712 Current Sensor	1	Current measurement
ZMPT101B Voltage Sensor	1	Voltage measurement
OLED SSD1306 Display	1	Real-time display
2-Channel Relay Module	1	Appliance switching
Bulb Holder	1	Bulb connection
240V Bulb	1	AC load
Fan	1	Appliance control
Jumper Wires	Multiple	Connections
Breadboard	1	Prototyping
WiFi Router	1	Internet connection
Smartphone	1	Blynk & Google Assistant
Buzzer (Optional)	1	Alerts
Power Supply	1	System power

8. Complete Wiring and Configuration

OLED Connections

OLED Pin	ESP32 Pin
-----	-----
VCC	3.3V
GND	GND
SDA	GPIO21
SCL	GPIO22

ACS712 Connections

ACS712 Pin	ESP32 Pin
-----	-----
VCC	5V
GND	GND
OUT	GPIO34

ZMPT101B Connections

ZMPT101B Pin	ESP32 Pin
-----	-----
VCC	5V
GND	GND
OUT	GPIO35

Relay Connections

Relay Pin	ESP32 Pin
-----	-----
IN1	GPIO26
IN2	GPIO27
VCC	5V
GND	GND

Appliance Connections

- * Relay Channel 1 → Bulb
- * Relay Channel 2 → Fan

9. Pin Connections

Device	ESP32 GPIO
-----	-----
ACS712	GPIO34
ZMPT101B	GPIO35
Relay 1	GPIO26
Relay 2	GPIO27
OLED SDA	GPIO21
OLED SCL	GPIO22

10. Circuit Diagram Explanation

The ESP32 acts as the central controller connected to sensors, relays, OLED, and IoT platforms.

The ACS712 measures current flow through the load.

The ZMPT101B measures AC voltage.

The relay module controls AC appliances.

The OLED display shows real-time values.

The ESP32 connects to WiFi and uploads sensor data to Blynk cloud.

Google Assistant commands are received via Sinric Pro and processed by ESP32.

11. Working Principle

1. ESP32 powers ON.
2. ESP32 connects to WiFi.
3. Sensors continuously monitor electrical parameters.
4. ESP32 calculates power.
5. OLED displays real-time values.
6. Data is uploaded to Blynk.
7. User controls appliances using:

- * Blynk App
- * Google Assistant

8. Relay switches ON/OFF appliances.

12. Step-by-Step Procedure

Hardware Setup

1. Connect ESP32.
2. Connect OLED display.
3. Connect ACS712.
4. Connect ZMPT101B.
5. Connect relay module.
6. Connect bulb and fan.
7. Verify AC wiring carefully.

Software Setup

1. Install Arduino IDE.
2. Install ESP32 board package.
3. Install required libraries.
4. Create Blynk account.
5. Create Sinric Pro account.
6. Configure Google Home.
7. Upload code.
8. Test IoT control.

14. Blynk Setup Guide

Steps

1. Install Blynk IoT app
2. Create template
3. Create datastreams:

| Datastream | Virtual Pin |

	-----		-----	
	Voltage		V0	
	Current		V1	
	Power		V2	
	Bulb Relay		V3	
	Fan Relay		V4	

4. Add widgets:

- * Gauge
- * Switch
- * SuperChart
- * Label

5. Copy:

- * Template ID
- * Device Name
- * Auth Token

6. Paste into Arduino code.

15. Sinric Pro Setup Guide

Steps

1. Create Sinric Pro account
2. Add Smart Switch device
3. Copy:

- * APP KEY
- * APP SECRET
- * Device ID

4. Integrate into ESP32 code.

16. Google Home Integration

Steps

1. Open Google Home app
2. Add device
3. Select Sinric Pro
4. Login to Sinric account
5. Devices appear automatically

Voice Commands

- * Hey Google, turn ON bulb
- * Hey Google, turn OFF fan

17. Features List

- * Real-time energy monitoring
- * OLED display visualization
- * Wireless appliance control
- * Blynk cloud monitoring
- * Google Assistant control

- * Relay automation
- * WiFi connectivity
- * SuperChart analytics
- * Smart home integration
- * Embedded IoT system

18. Applications

- * Smart homes
- * Industrial monitoring
- * Smart classrooms
- * Hostel energy monitoring
- * Smart laboratories
- * Energy-saving systems
- * Home automation

19. Advantages

- * Wireless operation
- * Remote monitoring
- * Energy-saving capability
- * Easy automation
- * Cloud integration
- * Voice assistant support
- * Low-cost implementation
- * Real-time analytics

20. Limitations

- * Requires internet connection
- * Depends on WiFi stability
- * Limited relay load capacity
- * Sensor calibration required

21. Future Improvements

- * AI-based energy prediction
- * Mobile notifications
- * Smart scheduling
- * Solar integration
- * Automatic overload protection
- * Face recognition control
- * MQTT integration
- * Machine learning analytics

22. Result

The developed system successfully monitored voltage, current, and power consumption while enabling wireless appliance control through Blynk and Google Assistant.

The system achieved stable IoT communication, real-time data visualization, and smart relay-based automation.

23. Conclusion

The project successfully demonstrates the implementation of an IoT-based Smart Energy Monitoring and Home Automation System using ESP32.

The integration of embedded systems, cloud IoT, relay automation, and voice assistant control provides an efficient, scalable, and intelligent smart home solution.
